European Commission

Joint Research Centre

EU food system monitoring framework

Fifteen new indicators to achieve a more balanced indicator coverage

Tóth, K., Acs, S., Barbero Vignola, G., Bosco, S., Caivano, A., Catarino, R., Druon, J.-N., De Laurentiis, V., De Jong, B., Ermolli, M., Galimberti, F., Guerrero, I., Gurria, P., Leite, J., Listorti, G., Lasarte Lopez, J., M'Barek, R., Puerta Pinero, C., Smallenbroek, O., Tamosiunas, S., Wollgast, J., Proietti, I.

2025



This document is a publication by the Joint Research Centre (JRC), the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The contents of this publication do not necessarily reflect the position or opinion of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of this publication. For information on the methodology and quality underlying the data used in this publication for which the source is neither Eurostat nor other Commission services, users should contact the referenced source. The designations employed and the presentation of material on the maps do not imply the expression of any opinion whatsoever on the part of the European Union concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

Contact information
Email: jrc-EU-FS-MONITORING@ec.europa.eu

The Joint Research Centre: EU Science Hub
https://joint-research-centre.ec.europa.eu

JRC143237
EUR 40522

PDF ISBN 978-92-68-33320-4 ISSN 1831-9424 doi:10.2760/5463127 KJ-01-25-556-EN-N

Luxembourg: Publications Office of the European Union, 2025

© European Union, 2025



The reuse policy of the European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (https://creativecommons.org/licenses/by/4.0/). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of photos or other material that is not owned by the European Union permission must be sought directly from the copyright holders.

Cover page illustration, © Päivi Sund

How to cite this report: Tóth, K., Acs, S., Barbero Vignola, G., Bosco, S., Caivano, A. et al., *EU Food system monitoring framework - Fifteen new indicators to achieve a more balanced indicator coverage,* Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/5463127, JRC143237.

Contents

Abstract 2

Foreword 3

Acknowledgements 4

1 Introduction 5

2 First upgrade of the dashboard 8

2.1 Comparison of the dashboard between the first and second release 9

2.1.1 Distribution of indicators among the sustainability dimensions 9

2.1.2 Distribution of indicators across the different food chain components 10

3 Development of new indicators 12

3.1 General principles 12

3.2 Indicators of the environmental dimension 14

3.2.1 Implemented modifications 15

3.2.2 Future development 16

3.3 Indicators of the economic dimension 18

3.3.1 Implemented modifications 18

3.3.2 Future development 21

3.4 Indicators of the social dimension 22

3.4.1 Implemented modifications 22

3.4.2 Future development 26

3.5 Indicators of the horizontal thematic areas 27

4 Health check of the system 28

4.1.1 Data preparation 28

4.1.2 Principal component analysis 30

4.1.3 Current health and future directions 33

5 Conclusion 34

References 35

List of abbreviations and definitions 37

List of figures 38

List of tables 39

Annexes 40

Annex 1. Recapitulative list of all indicators in the EU food system monitoring dashboard 40

Abstract

Food systems play a pivotal role in preserving the balance between environment and human activities. Their sustainability is key for operating within the planetary boundaries. The Joint Research Centre (JRC) published its first food system monitoring dashboard in November 2024, to provide information on progress towards a fair, healthy and environmentally friendly food system. In addition to selecting indicators for monitoring, the related work identified important knowledge gaps on food system sustainability. These gaps were marked with placeholders indicating research or data collection needs.

This report describes the first phase of the evolutionary maintenance of the dashboard, providing overall context and methodological steps for turning placeholders into headline or secondary indicators. Importantly, the report also informs on how data in the monitoring system is managed, ensuring balanced and non-repetitive coverage of the environmental, economic and social dimensions, as well as the components of the food supply chain. The first phase of placeholder development yielded 15 new indicators in the dashboard- out of which 9 had been formulated as placeholders in the November 2024 report while 6 were newly proposed and developed indicators. These indicators draw on newly discovered public data, data derived from original research, and data resulting from the combination of public and commercial data sources.

Beyond informing on the sustainability of the EU food system, the monitoring framework including the upgraded dashboard and this report, may also contribute to policy making on issues such as competitiveness, social inclusion and environment.

Foreword

The EU food system monitoring dashboard was published on 26 November 2024, accompanied by a technical report that describes the technical infrastructure (DataM platform), the conceptual framework and the methodology for indicator selection. The report also documents the indicators proposed for the first edition of the dashboard and highlights knowledge gaps in the form of placeholder indicators.

To close the knowledge gaps, the European Commission has decided to continue working to upgrade the dashboard between 2025 and 2026. This report informs on how the EU food system monitoring framework and dashboard indicators have been developed since November 2024, focusing on their new features.

This report maintains consistency with the previous report, as the underlying methodology, framework, and technical infrastructure remain unchanged. As a result, this report follows the same structure, layout and notation used in the first report.

Acknowledgements

The authors would like to acknowledge the input of Monica Crippa, (JRC C.5) and the internal reviewers, Marta Cubria Radio (JRC A.2) and Fabrizio Larcher (JRC F.5) for their valuable suggestions to improve the report. We thank our colleagues for the comments received during the consultations with the policy DGs. Last, but not least, we are grateful for the feedback of the Expert Group on General Food Law and Food System Sustainability and the Advisory group on Sustainability of Food Systems.

Authors

Tóth, Katalin
Ács, Szvetlana
Barbero Vignola, Giulia
Bosco Simona
Caivano, Arnaldo
Catarino, Rui
Druon, Jean-Noël
De Laurentiis, Valeria
De Jong, Beyhan
Ermolli, Monica
Galimberti Francesco
Guerrero, Irene
Gurría, Patricia
Leite, João
Listorti, Giulia
Lasarte Lopez, Jesus
M'barek, Robert
Puerta-Piñero, Carolina
Smallenbroek, Oscar
Tamosiunas, Saulius
Wollgast, Jan
Proietti, Ilaria

1 Introduction

The European Green Deal (EGD) proposes a new and inclusive growth strategy that highlights the need for a holistic and cross-sectoral approach, which includes climate, environment, agriculture and forestry, fisheries and aquaculture, energy, transport, industry and sustainable finance ([1]) (Tóth et al., 2024). The Farm to fork strategy (European Commission, 2020) was the first attempt to introduce a more integrated EU food policy (Schebesta & Candel, 2020), in which the development of an EU food system monitoring framework (FSMF) is highlighted as a significant opportunity for policy evaluation (Acs et al., 2025). The Vision for Agriculture and Food (European Commission, 2025) also highlights the need for a regular reporting on the progress regarding the successful delivery of the initiatives stated in the vision, including the possibility, following a proper assessment, to extend the on-farm sustainability assessments to the whole agrifood sector.

The European Commission's political guidelines 2024–2029 urge to implement the European Green Deal (EGD) during the current mandate. The EGD proposed a groundbreaking growth strategy that underscores the need for a cross-sectoral approach, encompassing climate, environment, agriculture and forestry, fisheries and aquaculture, energy, transport, industry, and sustainable finance (Tóth et al., 2024). Building on this, the Farm to Fork strategy (European Commission, 2020) has paved the way for a more integrated EU food policy and a systems view, with the development of an EU food system monitoring framework (FSMF). This tool has been designed to support policy-making by evaluating existing policies, informing new policy design, tracking progress towards specific goals, identifying trends and patterns, and comparing performance across different regions and sectors (Acs et al., 2025). Furthermore, the recently presented Vision for the future of agriculture and food (European Commission, 2025) highlights the importance of regular progress reporting on the initiatives outlined, including the potential expansion of on-farm sustainability assessments to the entire agrifood sector. By aligning these initiatives, the European Commission aims to create a sustainable food system that supports the EU's climate and environmental objectives, while promoting economic growth and social inclusivity.

The Joint Research Centre (JRC), in collaboration with DG SANTE, AGRI, MARE and ENV, as well as with involvement of other DGs of the European Commission (EC), experts delegated by the Member states in the Expert Group on General Food Law and Sustainability of Food Systems, and representatives of stakeholders in the Advisory Group on Sustainability of Food Systems, elaborated a monitoring framework and an associated dashboard to provide information on the environmental, economic and social sustainability aspects of Food system (FS). The report describing the framework and the related dashboard were first published on 26 November 2024.

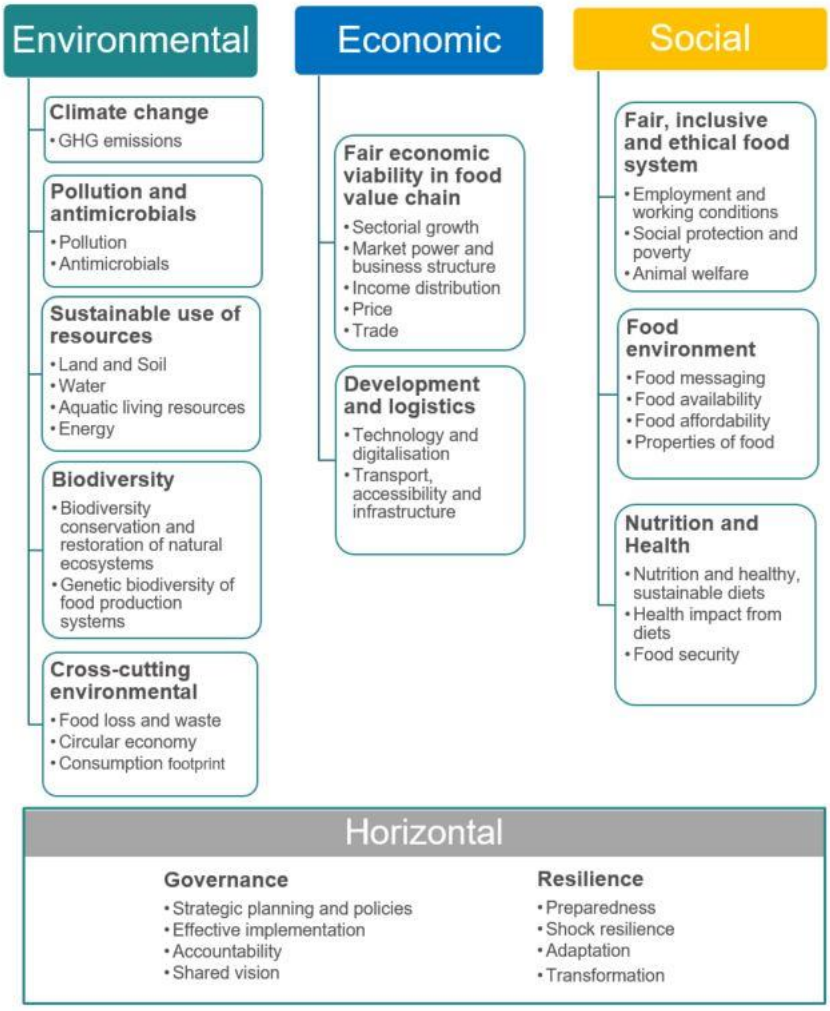
This framework is based on a conceptual model that includes the components of the food supply chain, such as primary production, food processing, distribution and consumption. The sustainability aspects of this model are hierarchically grouped in dimensions, thematic areas and domains as shown in **Figure 1**. This model serves for anchoring the indicators and assessing their thematic coverage in terms of relevance. The temporal characteristics, geographic coverage and quality of the methodology underpinning the indicators were evaluated based on harmonised metadata and the quality assessment framework, both documented in the first report.

([1]) https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en.

Reusing existing indicators and data sources was the driving approach in the development of the EU food system monitoring framework. This is a generally accepted good practice also underlined in the context of Sustainable Development Goals (SDG) monitoring: 'Official statistics derived from surveys and other official administrative data will play a critical, preeminent role. They will be complemented by unofficial data, and other performance metrics including business metrics, polling data, and georeferenced information on government facilities, among others (UN SDSN, 2015). Indeed, our first steps were focused on data delivered by public organisations and on highlighting the knowledge gaps. To mark areas of knowledge gaps where future research is needed, the concept of placeholder indicators was introduced. Placeholders are defined as conceptually important indicators for which data collection and/or conceptualisation is lacking.

The placeholder indicators, once ready to be implemented and after the scrutiny by the stakeholders' communities, can be adopted as headline or secondary indicators and displayed in the EU food system monitoring dashboard. However, when even the most thorough research does not yield satisfactory results for implementing a placeholder, the corresponding indicator is put in the 'pool' category of the JRC database until new methodological improvements or new data become available. It should be noted that in this first phase of development, no placeholder indicators were assigned to the pool. Nonetheless, the list of placeholders was actively maintained.

Figure 1. Food system sustainability model of the EU food system monitoring framework



Source: Own elaboration

The primary list of placeholders came from JRC own research and was extended based on the consultation with the services of the EC, the experts of the Advisory group on Sustainability of Food Systems, and the Member States experts in the Expert Group on General Food Law and Sustainability of Food Systems. The scope of the work described in this report is to populate placeholders with data, as well as to propose new indicator concepts in line with the evolving scientific findings and policy developments. The implemented placeholders will be upgraded to headline or secondary indicators, to reflect their relevance to food system sustainability.

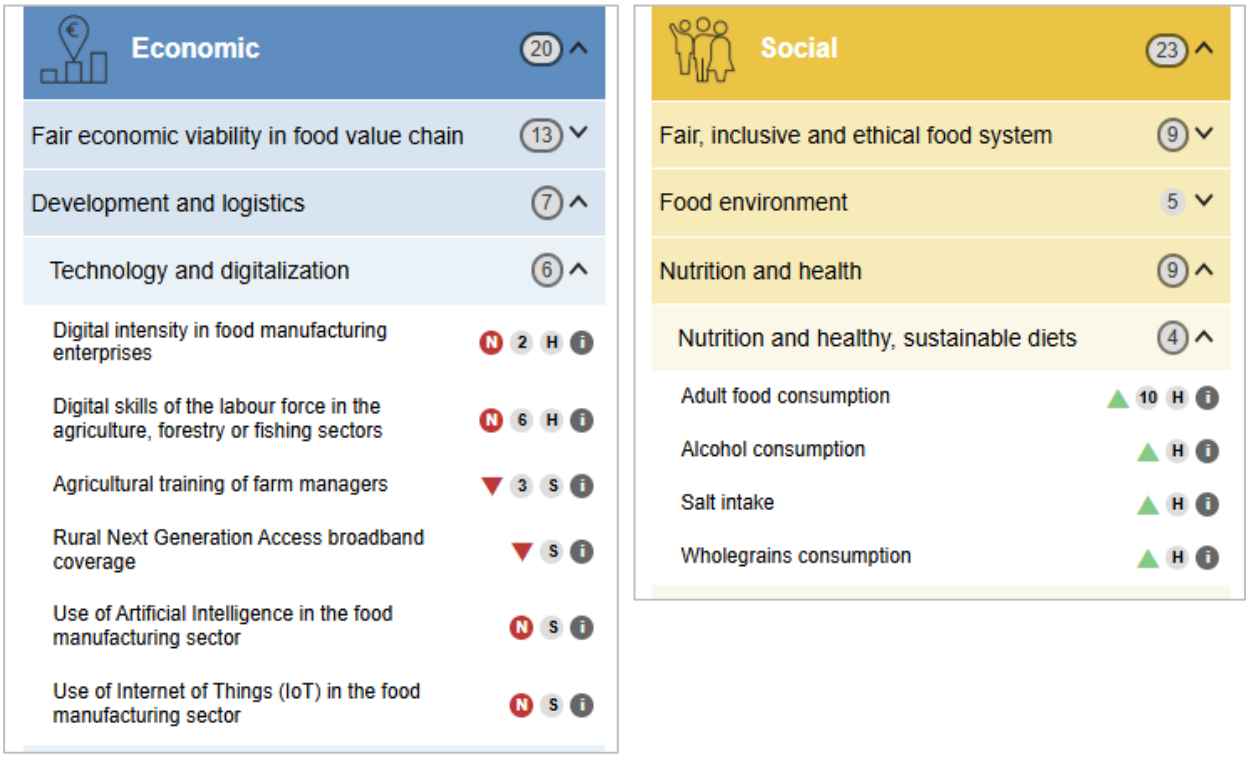
The JRC is carrying out work on placeholder implementation over 2025-2026, resulting in three upgrade releases of the dashboard. This report summarises the work in the first phase of placeholder development leading to the first upgrade of the EU food system monitoring dashboard.

2 First upgrade of the dashboard

The launch version of the dashboard, published on 26 November 2024, included only headline indicators that had been selected to ensure even coverage across the components of the food supply chain and the three sustainability dimensions. However, the accompanying technical report presented more indicators categorised as secondary, following consultations with Commission services, experts delegated by the Member States in the Expert Group on General Food Law and Sustainability of Food Systems, and representatives of stakeholders in the Advisory Group on Sustainability of Food Systems. These secondary indicators provided a more nuanced picture about the EU food system; however, their distribution was uneven within the system. In addition to covering the knowledge gaps, the new indicators introduced in the dashboard are also expected to improve the balance across the dimensions and the components of the food supply chain.

The first upgrade of the dashboard, which is described by the current report, publishes the 15 new indicators developed since 26 November 2024. In addition, the secondary indicators presented in the first technical report are also included, to achieve an overall more comprehensive and balanced coverage across the dimensions. The different categories of indicators, along with any relevant, reclassification, have an appropriate notation in the beta version of the dashboard, as shown in **Figure 2**. The dashboard indicates whether the indicators are new, compared to those included in the first technical report, using a red-circled letter "N", and whether the category of the indicator has changed with respect to the first version of the dashboard. The green triangle marks upgrades, while the red triangle stands for downgrades. These changes are explained in section 3.2.1.

Figure 2. Notation of changes in the dashboard



NB. H– headline, S– secondary, N– new, ▼– downgraded, ▲– upgraded indicator. Number of indicators dimensions, thematic areas and domains are indicated for each hierarchy level of the FS model. Number of sub-indicators (if any) are indicated for the indicators.

Source: Own elaboration

As explained under each section, the development of new indicators has been steered by the previously identified gaps (placeholder indicators) but also by recent policy developments as reflected in strategic documents published or announced by the European Commission (the Vision for Agriculture and Food, the Competitiveness Compass, the EU Plan on Cardiovascular Health etc.).

2.1 Comparison of the dashboard between the first and second release

2.1.1 Distribution of indicators among the sustainability dimensions

The purpose of upgrading the dashboard is to introduce new indicators in the system to address knowledge gaps on the sustainability of the EU food system. The knowledge gaps were mainly related to the domains in the economic and social dimensions, as well as to the components of the food supply chain beyond the primary production, i.e. in food processing, distribution and consumption. The initial collection of indicators was based on data, readily available from public sources. Therefore, areas governed by private sectors or demanding thorough research have remained unexplored. To provide a more comprehensive understanding of the EU food system, innovative approaches, new data sources, including market intelligence and cutting-edge technologies, such as earth observation and big data processing have been employed.

In this first phase of work, 15 new indicators were elaborated (out of which 9 had been formulated as placeholders in the November 2024 report, while 6 were newly developed indicators), resulting in changes to the dashboard composition as detailed in **Table 1**. This table shows how the number of indicators linked to the three sustainability dimensions has evolved from the first release of the dashboard to the second one, illustrating how the main efforts were focused on the Economic and Social dimensions, where more gaps had been identified.

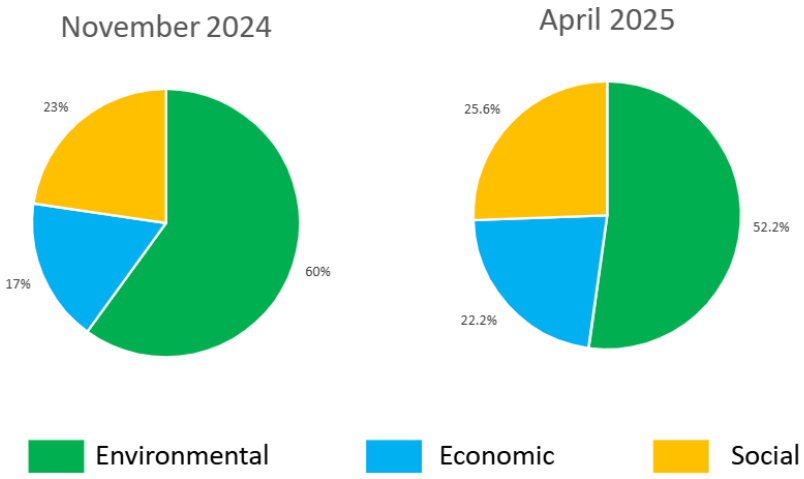
Table 1. Evolution of indicator numbers within the sustainability dimensions of the EU food system monitoring dashboard

	November 2024	**April 2025**	**Number of new indicators**
Environmental	45	47	2
Economic	13	20	7
Social	17	23	6
Total	**75**	**90**	**15**

Source: Own elaboration

The introduction of the new indicators improved the balance between the three thematic dimensions of the dashboard. The percentage changes between the dimensions are shown in **Figure 3**.

Figure 3. Proportions of the sustainability dimensions in the dashboard



Source: Own elaboration

However, analysing the internal structures of the dimensions reveals significant differences in the application of sub-indicators[2] (i.e. further categories within a specific indicator). In this respect, the Economic and Social dimensions follow a more similar approach. In the Economic dimensions, 11 out of the 20 indicators include sub-indicators, bringing 88 subcategories in the system. In the social dimension, six out of the 23 indicators, have a total of45 sub-indicators.

This practice is less common in the Environmental dimension, where only five out of the 47 indicators have further details, with a total of 18 sub-indicators. Applying sub-indicators in the Consumption footprint domain alone could reduce the number of indicators by 15, leaving one indicator for the single aggregated score and another for the detailed impact categories. The possibility for such intervention was also confirmed by the statistical analysis described in Section 4. Furthermore, the assignment of Circular economy and Food loss and waste to the Environmental dimension could be revisited. Harmonising the granularity of information within the indicators could substantially improve the perceived imbalance between the dimensions.

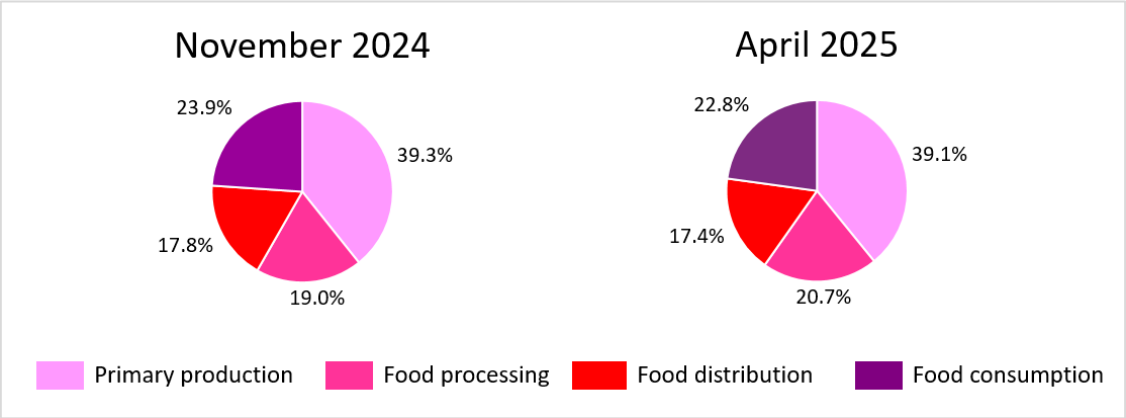
2.1.2 Distribution of indicators across the different food supply chain components

In relation to the coverage of different food chain components by indicators, it is important to recall that many indicators are not uniquely linked to one component but cover several parts of the food chain.

Having introduced seven specific indicators related to Food processing and four related to Food consumption, we expected a more significant change in the proportion between the components of the food supply chain. However, as shown in **Figure 4**, there was only a slight improvement in Food processing.

[2] For example, indicator 'Value Added along the food chain' has four sub-indicators (further categories), such as Agriculture, Fishing and aquaculture, Food, Beverages.

Figure 4. Proportion of food supply chain components in the dashboard



Source: Own elaboration

The reason lies in the nature of the indicators within each component. The numbers of indicators linked to the different components of the supply chain is presented in **Table 2**.

Table 2. Changes of indicator numbers within the supply chain components of the dashboard

	November 2024	**April 2025**	**Number of new indicators**
Primary production	64	72	8
Food processing	31	38	7
Food distribution	29	32	3
Food consumption	39	42	3
Total	163	184	19

NB. The total number of indicators does not equal to that in **Table 1**, as an indicator can be linked to more than one component of the food supply chain.

Source: Own elaboration

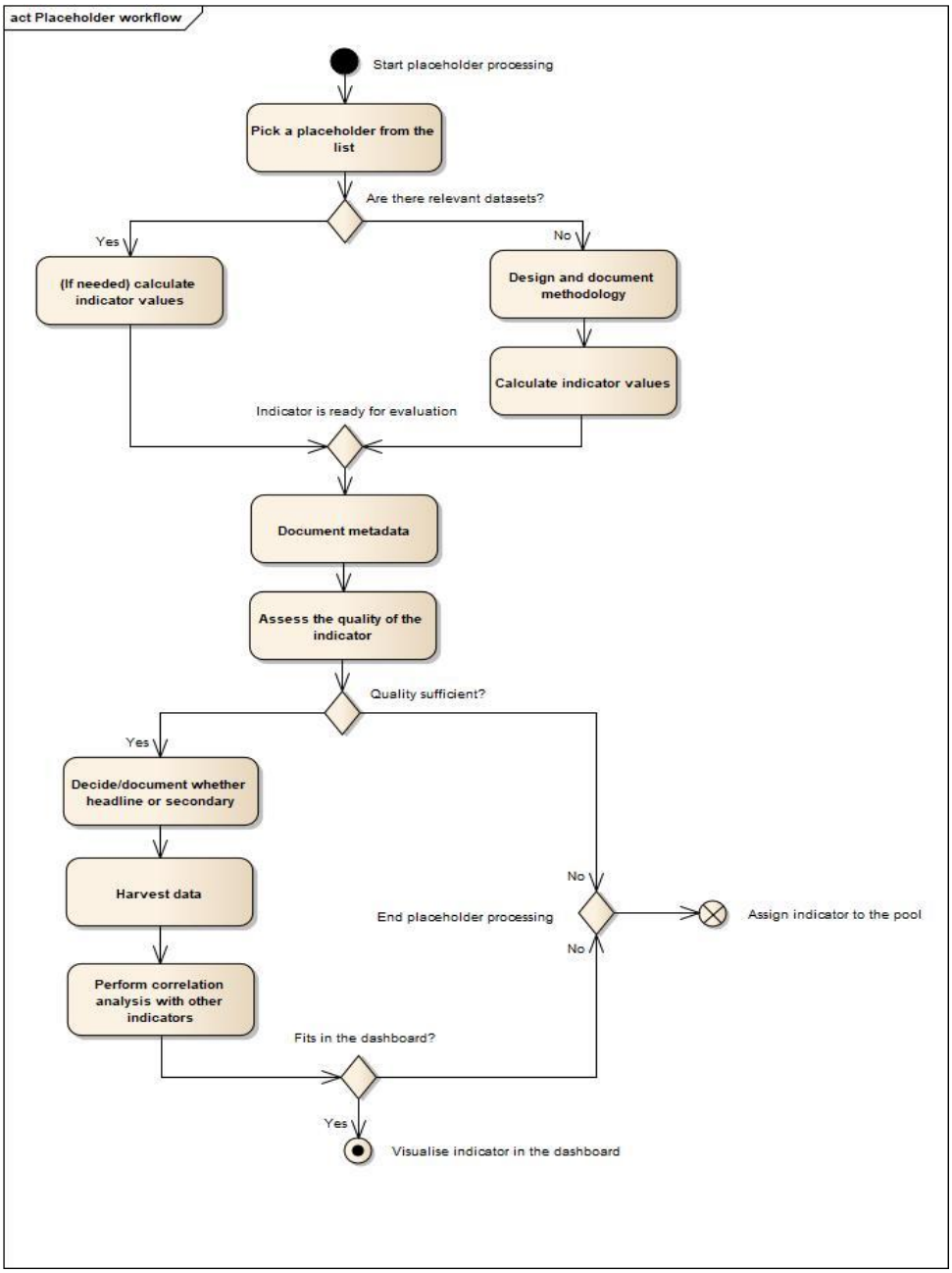
While in November 2024, a dashboard indicator was linked to an average of 2,2 supply chain components, this value was reduced in April 2025 to 2,0 which is not a significant change. This means that both the original set of the indicators and the newly added ones have very few specific indicators for the individual components of the supply chain. Instead, the majority of indicators provide value for multiple components, often in the form of aggregates. This situation is particularly striking for Food distribution, where, across all dimensions, only nine indicators provide aspects that are unique to this component and not relevant to the rest of the framework.

3 Development of new indicators

3.1 General principles

The methodology and the workflow for indicator development have not changed since the first publication of the dashboard. Consequently, the major processing steps for developing new indicators from placeholders are the same as those reported in the first technical report and as they are illustrated in **Figure 5**.

Figure 5. Steps of placeholder processing



Source: own elaboration

Since the list of placeholder indicators was compiled after a preliminary analysis of their relevance, the first important step was to verify the existence of datasets to implement them. In some cases, we managed to identify datasets that directly provided suitable indicators or allowed their calculation in a relatively simple way. In other cases, indicators were derived from our own research, which involved a detailed documentation of the applied methodology.

The placeholder indicators that passed the quality assessment process were proposed for the dashboard. The gradually increasing number of indicators in the dashboard created the risk of (partial) repetitions or losing the focus. To avoid this, an overall correlation analysis between the indicators was added to the workflow, as described in section 4.1.2. The final step was to reassess the indicators selected for the dashboard within each domain and the dashboard as a whole. This resulted in applying the necessary changes in the category of the indicators as described below:

- **Adding a new headline or secondary indicator** as compared to the first publication of the dashboard. Such an indicator can **result from upgrading** a previously identified placeholder to a headline or secondary indicator **or** from the discovery of new indicators during **new research**.
- **Downgrading a headline indicator to a secondary**. This action was taken when a newly developed indicator provided broader, or more relevant information within the scope of a specific domain.
- **Downgrading an indicator to the pool**. This occurred when a placeholder indicator does not appear implementable, or a secondary indicator can be replaced with a more robust one based on our quality assessment framework.
- **Extending the scope of an indicator** was applied to provide indicator values for one or more additional supply chain components than proposed in the first edition.

To better reflect the thematic content of the implemented placeholders, changing or fine-tuning their names might have been necessary. An overview of all changes occurred in the first phase of placeholder development is provided **Table 3**.

Table 3. New and updated indicators elaborated during the first upgrade release of the EU Food System Monitoring Framework

Indicator	Domain	Category (and change)
Newly introduced indicators		
Food losses	Food loss and waste	Headline
Digital skills of the labour force in the agricultural, forestry or fishing sectors	Technology and digitalisation	Headline
Digital intensity in food manufacturing enterprises	Technology and digitalisation	Headline
Use of Artificial Intelligence in the food manufacturing sector	Technology and digitalisation	Secondary
Use of Internet of Things (IoT) in the food manufacturing sector	Technology and digitalisation	Secondary
Job vacancies in the agrifood sector	Employment	Secondary

Upgraded indicators		
Generation and use of by-products	Circular economy	Headline (corresponds to “Circular economy” ex-placeholder)
Average salary per sector	Income distribution	Headline (ex-placeholder)
Employee earnings ratio	Income distribution	Secondary (ex-placeholder)
Labour income share	Income distribution	Secondary (corresponds to "Employee remuneration as a share of value added, by sector" ex-placeholder)
Gender employment gap in the food sector	Employment	Secondary (ex-placeholder)
Adult consumption by food groups	Nutrition and healthy, sustainable diets	Headline (corresponds to “Food consumption (food groups and other dietary factors)” ex-placeholder)
Salt intake	Nutrition and healthy, sustainable diets	Headline (partial implementation of “Food consumption (food groups and other dietary factors)” ex-placeholder)
Alcohol consumption	Nutrition and healthy, sustainable diets	Headline (partial implementation of “Food consumption (food groups and other dietary factors)” ex-placeholder)
Wholegrains consumption	Nutrition and healthy, sustainable diets	Headline (partial implementation of “Food consumption (food groups and other dietary factors)” ex-placeholder)
Downgraded indicators		
Agricultural training of farm managers	Technology and digitalisation	Secondary (ex-headline)
Rural Next Generation Access broadband coverage	Technology and digitalisation	Secondary (ex-headline)
Extended indicators		
Value added along the food chain	Sectoral growth	Headline

Source: Own elaboration

The following subsections detail the changes in the dashboard as compared to its first edition in November 2024. In addition to showing the scope and role of the indicators and the associated changes, we also explain the rationale behind these modifications.

3.2 Indicators of the environmental dimension

The environmental dimension consists of five thematic areas: Climate change, Pollution and antimicrobial, Sustainable use of resources, Biodiversity, and Cross-cutting environmental. Thematic areas are further broken down into domains.

3.2.1 Implemented modifications

In the Environmental dimension, modifications have been performed primarily in the **Cross-cutting thematic area,** with an additional adjustment introduced in the **Pollution and antimicrobials** thematic area. The major changes include the introduction of two headlines in the Cross-cutting thematic area: one in the **Food loss and waste** domain and the other in the **Circular economy** domain. In Food loss and waste domain, we introduced the **Food losses** indicator as a headline, and we promoted the **Generation and use of by-products** in the Circular economy domain from a placeholder to headline indicator. The overview of these changes is included in **Table 4**.

Table 4. New and updated indicators for Cross-cutting environmental thematic area.

Domain	Indicator	PP	FP	FD	FC
Food loss and waste	Food losses				
Circular economy	Generation and use of by-products	↑	↑		

PP-primary production, FP -Food processing, FD -Food distribution, FC -Food consumption; blue– headline, ↑– upgraded indicator.

Source: Own elaboration

The **Food losses** indicator was designed to quantify and monitor food losses occurring at primary production. It stems from the Food System Material Flow Analysis (FS-MFA) model[3], which was built on yearly material flow analyses of the food system within every Member State. This indicator focuses on losses of crops prior to harvesting (including crops ploughed in or left on the field) and mortality of animals before slaughter. This indicator complements the quantification of food waste at primary production, as described in De Laurentiis et al. (2024).

The Food losses indicator is provided at food group level, such as sugar, cereals, fruit and nuts, vegetables, potatoes, oilseeds, meat, fish, eggs, and dairy. Alongside Food waste, the other headline indicator already in the Dashboard, this indicator supports the EU's efforts to meet SDG 12.3, which focuses on halving per capita food waste and reducing food losses by 2030. Reducing food waste remains at the forefront of the EU agenda, with the fight against food loss and waste continuing to be a top priority in the years to come. The recent revision of the Waste Framework Directive[4] has reinforced this commitment, and the Vision for Agriculture and Food has also underscored the importance of tackling this critical issue. Reducing and valuing food losses and food waste will not only benefit EU citizens, farmers and all other actors across the food supply chain but will also increase the sustainability of the EU food system, contributing to more efficient resource use and food security.

The other headline indicator, **Generation and use of by-products**, was a placeholder (named "Circular economy") that has been upgraded to headline, as it represents a significant contribution to circular economy. Indeed, the analysis of generation and use of by-products in the food supply chain supports the assessment of the circularity of food systems, in line with the objectives of the

[3] https://web.jrc.ec.europa.eu/policy-model-inventory/explore/models/model-fs-mfa/

[4] https://ec.europa.eu/commission/presscorner/detail/en/ip_25_548

circular economy action plan (CEAP)[5], one of the main building blocks of the EGD. Indicators on the circular economy are particularly relevant and timely given that circularity remains at top of the policy agenda, as recognised in the Vision for Agriculture and Food, which emphasises the potential that bioeconomy and circularity offer for agriculture, forestry and the entire food system, as well as for reducing our critical dependencies. The new Bioeconomy Strategy, to be presented by the end of 2025, will aim among other elements at accelerating the commercialisation of bio-based and circular solutions. This will be particularly beneficial for the farming community by enabling diversification of value streams, valorisation of farm residues, strengthening the role of primary producers in the value chain and generating new jobs in the rural areas.

Similarly to the Food loss indicator, the Generation and use of by-products is based on the FS-MFA model. It is designed to quantify and monitor by products generated at primary production and food processing, which are used then to produce animal feed and non-food goods. Like the indicator for Food losses, the values of this indicator are provided at food group level, adding the "cocoa and coffee" category. The indicator is divided into two sub-indicators, one referring to the by-products generated at primary production, and the second to the by-products generated at processing.

The two indicators were obtained by combining statistical data on crops and livestock production and trade with coefficients taken from the scientific and technical literature, expressing rates of food losses, food waste and by-products generation specific for the stage of the food supply chain, the food group, and, when possible, the country of production, as well as data collected from trade associations in some cases. Details on the methodology and input data used to derive both indicators are provided in De Laurentiis et al. (2021, 2023, 2024).

A minor adjustment has also been made in the Pollution and Antimicrobial thematic area. Under the Pollution domain, the two pesticide-related indicators previously titled "Use and risk of chemical pesticides (F2F pesticide reduction target 1)" and "Use of the more hazardous pesticides (F2F pesticide reduction target 2)" have been renamed to "Use and risk of chemical pesticides" and "Use of the more hazardous pesticides", respectively, removing references to the F2F reduction targets.

3.2.2 Future development

Despite the relatively high number of indicators in the Environmental dimension further work is foreseen to provide information on several new aspects of particular interest, as included in **Table 5**.

Table 5. Overview of expected placeholder indicator development in the Environmental dimension

Thematic area	**Domain**	**Placeholder indicator**	**Work planned**	
Climate change		GHG emissions	GHG food system emissions	Disaggregation according to the components of the food supply chain.

[5] https://environment.ec.europa.eu/strategy/circular-economy-action-plan_en

Thematic area	Domain	Placeholder indicator	Work planned	
			GHG emissions from livestock	New area of research. Quantification of emission from livestock from the GHG inventory.
Pollution and antimicrobials		Pollution	Pesticides in rivers, lakes and groundwater in Europe	New area of research. Aggregation of input data at the country level.
		Antimicrobials	Antimicrobial resistance from manure to soil and/or surface water	New area of research. Feasibility study for soil; separation of livestock and human borne pollution.
Sustainable use of resources		Land and soil	Crop diversity	Share of agricultural land complying to good practices of crop diversification
			Total biomass supply for food purposes, including inputs	New area of research. Adopting the indicator from the EU Bioeconomy monitoring system to EU FSMF.
			Drought impact on agriculture	Deriving the indicator from remote sensing data.
			Global deforestation index due to EU food consumption	Deriving the indicator based on deforestation data and LCA.
			Soil sealing in agricultural area	Deriving the indicator based on remote sensing data or subsetting the soil imperviousness indicator of the EEA for agricultural land.
			Agricultural land covered with landscape features	Deriving the indicator based on farmers declarations and/or remote sensing.
		Water	Abstraction of irrigation water, Use of rainwater; water storage/collection	New area of research in line with the Water resilience strategy.
		Aquatic living resources	Number of assessed fish stocks for which	This indicator was developed in the Mediterranean Sea in 2024

Thematic area	Domain	Placeholder indicator	Work planned	
			fishing mortality (F) is above/below FMSY	and will therefore naturally substitute the current headline indicator on this topic.
			Share of sustainable aquaculture	New area of research. Disaggregating aquaculture production data based on trophic levels.
		Energy	Renewable energy in different food system sectors	New area of research. Estimation of the use/production of renewable energy in different components of the food supply chain.
Biodiversity		Biodiversity conservation and restoration of natural ecosystems	Pressure by invasive alien species on ecosystems	Deriving country values from the EASIN database[6]
			Impact of fisheries on marine biodiversity	Working on the evaluation of fishing gears to promote best practices and monitor progress.

Source: Own elaboration

3.3 Indicators of the economic dimension

The economic dimension consists of two thematic areas: **Fair economic viability in food value chain** and **Development and logistics**. Both areas have been updated by new indicators.

3.3.1 Implemented modifications

The changes in this dimension include

- Promoting placeholders to headline or secondary indicators,
- Extending the coverage of existing indicators to more components of the food supply chain, and
- Demoting a previously published indicator from headline to secondary.

In the **Fair economic viability in food value chain** thematic area, changes only took place for the **Sectoral growth** and **Income distribution** domains as shown in **Table 6**. Indicators under

[6] https://easin.jrc.ec.europa.eu/easin

this section aim to contribute to inform policy thinking on the topic of a fair and equitable food chain, which is one of the areas of focus set out by the Vision for Agriculture and Food.

Table 6. New and updated indicators for Fair economic viability in food value chain

Domain	Indicator	PP	FP	FD	FC
Sectoral growth	Value added along the food chain				
Income distribution	Average salary per sector	↑	↑	↑	
	Employee earnings ratio	↑	↑	↑	
	Labour income share	↑	↑	↑	

PP-primary production, FP -Food processing, FD -Food distribution, FC -Food consumption; blue– headline, green– secondary, ↑– upgraded indicator.

Source: Own elaboration

The indicator **Value added along the food chain** was one of the headline indicators published with the first release of the dashboard. However, it covered the food chain partially as the trade and services sectors were not reported. These segments have now been added to provide a more comprehensive view. This extension contributes to a more complete coverage of the food value chain.

The three placeholder indicators in the Income distribution domain have now been completed and promoted to either headline or secondary indicators. In addition to the existing headline indicator Farmers' income in agriculture compared with wages in the rest of the economy, the indicator **Average salary per sector** has now also been designated as headline indicator. It serves as a proxy for assessing the position of a specific sector within the overall income distribution. This indicator shows the mean salary earned by persons employed in the different sectors of the food system, covering the food processing and food distribution sectors alongside the primary food production (Lasarte López et. al., 2025). Alongside the indicator on value added, this indicator contributes to the understanding of the economic positions of different actors in the food chain and can therefore inform policy thinking on the key objective of a fair living standard in the agri-food sector.

The indicator **Employee earnings ratio** is proposed for the dashboard as a secondary indicator, providing an overview of the income distribution throughout the food chain and highlighting where employees are remunerated in relation to the value they add. It is calculated by the ratio between the average salary and the value added per worker per sector. **Labour income share** is also introduced as secondary indicator in this domain, aiming to monitor the ability of farmers and other food chain employees to capture a fair share of the added value. This indicator was initially a placeholder, with a name 'Employee remuneration as a share of value added, by sector' that was changed to 'Labour income share' to align with the source data. It measures the proportion of employee remuneration relative to the value added (Lasarte López et. al., 2025). These secondary indicators add further depth to the understanding of the economic positions of different actors in the food chain.

There have been no changes in the domains of **Market power and business structure, Price**, and **Trade**.

In the **Development and logistics** thematic area, the only updates were made to the Technology and digitisation thematic area, where four new indicators were added, as included in **Table 7**. As

digitalisation and diffusion of advanced technologies across the European economy are a key ingredient to boost productivity growth (as recognised by the Competitiveness Compass Communication[7]), these indicators add value by strengthening the understanding of the state of play and trends on these topics for the food system.

The Vision for Agriculture and Food also highlights that the digital transition can significantly enhance operations and drive innovation and revolutionise the way we produce food, taking care of the environment, climate and people. Nonetheless the adoption of digital tools lags behind in agriculture and other parts of the food system. Perceived high costs, lack of digital skills and trust, absence of tailored solutions and connectivity issues are among the main reasons why farmers are not fully tapping into the digitalisation wave. Therefore, indicators proposed for the dashboard are geared towards covering precisely these aspects: skills, connectivity, use of digital technologies.

Table 7. New and updated indicators for Development and logistics

Domain	Indicator	PP	FP	FD	FC
Technology and digitalisation	Agricultural training of farm managers	↓			
	Digital skills of the labour force in the agricultural, forestry or fishing sectors				
	Digital intensity in food manufacturing enterprises				
	Rural Next Generation Access broadband coverage	↓	↓	↓	↓
	Use of Artificial Intelligence in the food manufacturing sector				
	Use of Internet of Things (IoT) in the food manufacturing sector				

PP-primary production, FP -Food processing, FD -Food distribution, FC -Food consumption; blue– headline, green– secondary, ↓ -indicator downgraded from headline to secondary as compared to the first edition of the dashboard.

Source: Own elaboration

The headline indicators of the previous edition of the dashboard, the **Agricultural training of farm managers** and the **Rural next generation access broadband coverage** were reclassified as secondary indicators. This change was made because two of the new indicators provide a more comprehensive representation of the domain: Digital skills of the labour force in the agricultural, forestry or fishing sector, and Digital intensity in food manufacturing enterprises.

One of the new headline indicators **Digital skills of the labour force in the agricultural, forestry or fishing sector** measures the digital proficiency (e.g. internet or software use) of persons aged 16-74 employed in these sectors. The indicator is of importance as digital skills are a key performance metric for the labour force, particularly in the food production sectors, and digital transformation can support the EGD, as outlined in the Digital Decade framework (European Commission, 2023).

[7] https://commission.europa.eu/document/download/10017eb1-4722-4333-add2-e0ed18105a34_en

The other new headline indicator, **Digital intensity in food manufacturing enterprises** measures the proportion of enterprises in the food manufacturing sectors with high and very high digital intensity, based on the survey on ICT usage and e-commerce in enterprises. This indicator is also one of the key performance metrics in the context of the Digital Decade. Furthermore, the Digital Compass sets a 2030 target for more than 90% of EU small and medium-sized enterprises (SMEs) to achieve at least a basic level of digital intensity (European Commission, 2023). These two indicators are complementary in terms of supply chain stage coverage, as the first one focuses on primary production, while the second one covers the food manufacturing sector.

The Use of Artificial Intelligence in the food manufacturing sector and Use of Internet of Things (IoT) in the food manufacturing sector are classified as secondary indicators. The **Use of Artificial Intelligence in the food manufacturing sector** indicator measures the share of food and beverages manufacturing enterprises that utilise at least one of the AI technologies. On the other hand, the indicator **Use of Internet of Things (IoT) in the food manufacturing sector** estimates the proportion of enterprises active in food manufacturing that use IoT in their business activities (use of interconnected devices or systems by enterprises and self-employed persons). Both indicators facilitate the monitoring of the EU's digital targets for 2030, as set out in the Digital Compass for the EU's Digital Decade (European Commission, 2023).

In the domain of **Transport, accessibility and infrastructure**, there have been no changes.

3.3.2 Future development

As shown in **Table 8**, we plan to publish five additional indicators in the economic dimension for the second and third release of the dashboard. The current placeholder indicators Import dependency and Balassa index are scheduled for publication in the second release, while Market concentration, Fertiliser self-sufficiency rate, and Patent application in the agri-food sector are planned for the third release.

Table 8. Overview of expected placeholder indicator development in the Economic dimension

Thematic area	Domain	Placeholder indicator	Work planned
Fair economic viability in food value chain	Market power and business structure	Market concentration	Exploratory work ongoing.
	Trade	Fertiliser self-sufficiency rate	Identification of data sources.
		Import dependency	Data classification and quality check.
		Balassa Index	Exploratory work ongoing.
Development and logistics	Technology and digitalisation	Patent application in the agri-food sector	New area of research. Exploratory work ongoing.

Source: Own elaboration

3.4 Indicators of the social dimension

The social dimension of the EU FSMF encompasses three thematic areas: **Fair, inclusive and ethical food system**, **Food environment,** and **Nutrition and health**. Within this dimension, modifications have been implemented in the Fair, inclusive and ethical food system and Nutrition and health thematic areas.

3.4.1 Implemented modifications

In the **Fair, inclusive and ethical food system** thematic area, specifically within the **Employment** domain, two secondary indicators have been added to the existing headline indicators enriching the domain with additional information on the employment situation in the member states (MS). Concerning the **Nutrition and health** thematic area, adjustments have been made exclusively in the **Nutrition and healthy, sustainable diets** domain, where four indicators have been promoted from placeholders to headline indicators.

In the Fair, inclusive and ethical food system thematic area, changes only took place for the Employment domain. Within this domain, Gender employment gap in the food sector has been upgraded from a placeholder to a secondary indicator, and Job vacancies in the agrifood sector has been introduced as a new secondary indicator. These changes are included in **Table 9**.

Table 9. New and updated indicators for Fair, inclusive and ethical food system

Domain	Indicator	PP	FP	FD	FC
Employment	Gender employment gap in the food sector	↑	↑	↑	↑
	Job vacancies in the agrifood sector				

PP-primary production, FP -Food processing, FD -Food distribution, FC -Food consumption; green– secondary, ↑– upgraded from placeholder to secondary indicator as compared to the previous version of the framework.

Source: Own elaboration

The **Gender employment gap in the food sector** measures the difference between the employment rates of men and women aged 20 to 64. The indicator calculated here considers only a subset of economic activities, giving an overview of the gender equity in the food system and specifically regarding the gender employment gap. The following activities are included:

- Crop and animal production, hunting and related service activities,
- Fishing and aquaculture,
- Manufacture of food products and beverages and related service activities,
- Manufacture of tobacco products.

This indicator permits to compare female/male employment rates, in order to evaluate inclusion and gender equity in the food system. A similar indicator, applied to the whole sector of economic activities, is part of the SDG indicator set. It is used to monitor progress towards SDG 5 on gender equality. The indicator is also included as a headline indicator in the Social Scoreboard for the European

Pillar of Social Rights[8]. Increasing female labour market participation and the equal economic independence of women and men is also one of the priorities of the EU Gender Equality Strategy[9].

The **Job vacancies in the agrifood sector** measures the proportion of paid positions that are newly created, unoccupied, or about to become vacant, expressed as a percentage[10]. This indicator provides insights into the level and structure of labour demand. However, it doesn't measure the quality of jobs, occupational safety and health, social protection and wages that define the attractiveness and generational renewal potential of the sector. The job vacancy rate may partially reflect unmet labour demand and potential mismatches between the skills and availability of unemployed individuals and those sought by employers. However, the interpretation of this indicator is context-dependent: a positive trend may indicate sector growth or a shortage of workers, whereas a negative trend could reflect sufficient labour availability or a decline in sector development.

In the domain of **Social protection and poverty** no modifications have been performed so far. This is a crucial domain, but the challenge is the lack of harmonised data across the EU. This is a direct consequence of EU Member States having different arrangements (management and funding) for their social security systems and for contracting seasonal agricultural work. Even though ILOSTAT provides access to statistical information for EU countries, but unfortunately there is no disaggregation by the food/agriculture sector. For these reasons, further analysis and estimates are required.

For the **Animal Welfare**, no other indicators have been included due to the lack of data. This clearly highlights the need for harmonised data collection in the EU. The JRC invests further efforts to address this gap by appropriate proxies, at least.

Adequate monitoring of food consumption can inform on the progress towards healthy and sustainable diets, which is crucial for achieving sustainable food systems. The Global Burden of Disease (GBD) study estimates that, in the European Union in 2021, approximately 60,000 deaths and over 11,5 million disability-adjusted life years (DALYs)[11] are attributable to dietary risks (GBD, 2021). This disease burden corresponds exclusively to non-communicable diseases (NCDs) such as cardiovascular diseases, neoplasms, diabetes, and kidney diseases (Garcia Herrero et al., 2022; GBD, 2021).

In most European populations, the intake of saturated fatty acids (SFA), salt, and free sugars, exceeds current dietary recommendations, while dietary fibre intake falls below the recommended levels (European Commission, 2024). When comparing food group intakes to Food-Based Dietary Guidelines (FBDGs) in the EU, consumption of fruit, vegetables, legumes, whole grains, nuts, and seeds, as well as dairy products and fish, is lower, and red and processed meat, sugar-sweetened beverages, and alcohol intake is higher than recommended (Springmann et al., 2020).

The Healthier together – EU NCDs initiative launched by the Commission in 2021 and ongoing until 2027 is supporting EU countries to identify and implement effective policies and actions to reduce the burden of major NCDs and improve citizens' health and well-being. The initiatives highlight,

---

8 https://ec.europa.eu/eurostat/cache/dashboard/social-scoreboard/

9 https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/gender-equality/gender-equality-strategy_en

10 https://ec.europa.eu/eurostat/cache/metadata/en/jvs_esms.htm

11 https://knowledge4policy.ec.europa.eu/glossary-item/disability-adjusted-life-year_en

among other aspects, determinants that contribute to the development of the major NCDs: tobacco use, harmful consumption of alcohol, unhealthy diets and lack of physical inactivity[12]. Recently, the Commission has announced the intention to take action on the most important cause of death in the EU –cardiovascular disease – through a European Cardiovascular Health Plan[13,14]. Most cardiovascular diseases can be prevented by addressing risk factors, including dietary risk factors and diet-related risk factors such as obesity, high blood pressure, high blood glucose, high blood cholesterol.

Concerning the **Nutrition and healthy, sustainable diets** domain the first technical report included a single placeholder for the four indicators included in **Table 10**. We had to split the placeholder due to transparency and technical reasons and, as the four indicators have different data sources and data collection methodologies as explained in the following paragraphs.

Table 10. New and updated indicators for Nutrition and health

Domain	Indicator	PP	FP	FD	FC
Nutrition and healthy, sustainable diets	Adult consumption by food groups				
	Salt intake				↑
	Alcohol consumption				↑
	Wholegrains consumption				↑

PP-primary production, FP -Food processing, FD -Food distribution, FC -Food consumption; blue– headline, green– secondary, ↑ -upgraded indicator from placeholder to secondary, as compared of the first version of the framework.

Source: Own elaboration

Given the strong link between NCDs and dietary patterns, indicators covering food consumption can support health-related policies. Therefore, the JRC has developed a set of indicators on **Adult consumption by food groups** categorised as headline. The European Food Safety Agency (EFSA) provides reliable and representative dietary information from national dietary surveys across Europe, using harmonised data collection methods[15]. National dietary surveys are preferred sources of high-quality information on food consumption at consumer level. However, these surveys are not conducted regularly, limiting their effectiveness for continuous monitoring. In addition, survey years across EU countries range substantially, requiring a fit for purpose approach to adjust and produce updates estimates that may reflect changes in dietary behaviour in the population to recent years. To overcome this issue, we developed a method to estimate annual food consumption estimates for

12 https://health.ec.europa.eu/non-communicable-diseases/healthier-together-eu-non-communicable-diseases-initiative_en#latest-updates

13 https://commission.europa.eu/document/download/b1817a1b-e62e-4949-bbb8-ebf29b54c8bd_en?filename=Mission%20letter%20-%20VARHELYI.pdf

14 https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14755-EU-cardiovascular-health-plan_en

15 https://www.efsa.europa.eu/en/data-report/food-consumption-data#the-efsa-comprehensive-european-food-consumption-database

key dietary aspects by integrating data EFSA food consumption data with market sales data (Euromonitor).

The first step towards establishing this set of food consumption indicators was to identify food groups of interest according to international references (e.g. WHO, FAO, EAT-Lancet diet) and national food based dietary guidelines regarding what constitutes healthy, sustainable diets.

In the EFSA food consumption database, we have explored the FoodEx2 classification system to identify which dietary factors of interest for monitoring could be quantifiable from the database. The food groups selected were starchy, vegetables, fruit, legumes, nuts and seeds, fish and seafood, milk and dairy, eggs, poultry, red meat, processed meat, and sugar-sweetened beverages. Nutrient intakes were not included as the database provides data only at the food item level. We retrieved adult food consumption data (ages 18-64) in grams per day from 22 EU countries as available in the database.

To project updated consumption estimates, we applied annual percentage changes from Euromonitor[16] food sales data to adjust the baseline estimates for each food group. We analysed food sales data to determine the annual percentage change for each food group, by country. To derive updated food consumption estimates for 2024 across all countries, we considered the year of implementation of each national dietary survey as the baseline (time 0). We then applied the annual rate of change (%) from the corresponding food group sales data to update the EFSA consumption estimates.

This approach tracks annual changes in food consumption (g/person/day) across 22 EU countries up to 2024 for starchy, fruit, vegetables, legumes, nuts and seeds, fish, milk and dairy, eggs, red meat, processed meat and sugar sweetened beverages. The quality of these metrics lies in its sound methodology based on reliable and harmonised data collection methods as developed by the EFSA, ensuring accurate, up-to-date data that is more reflective of real-time consumption trends.

Excessive salt consumption, whether from industrial food processing, food preservation or discretionary sodium added during cooking or eating, is one of the leading dietary risk factors for non-communicable diseases. The GBD study estimated that, in 2019 a diet high in sodium was responsible for over 155,000 deaths and more than 2.5 million DALYs in the EU (European Commission, 2024). The EFSA has also reported a link between high sodium intake and an increased risk of morbidity and mortality from cardiovascular diseases, including coronary heart disease and stroke (EFSA, 2005). For this reason, monitoring **Salt intake** is highly relevant to reach healthy food systems. The necessary data come from the GBD estimate and are expressed as the average (mean) daily salt intake (in grams) among adults aged 25 and older (Chen et al., 2021).

According to the 2021 GBD study (GBD, 2021), almost 210,000 deaths and around 6.8 million DALYs in the EU Member States are estimated to be attributable to high alcohol consumption. Of these, approximately 190,000 deaths and over 5.8 million DALYs were due to NDCs, while the remaining cases to injuries. Among the NCDs, around 57,000 deaths were caused by cancer and another 50,000 by cirrhosis and chronic liver diseases (European Commission, 2024).

[16] https://www.euromonitor.com/

To address alcohol-related harm, several policies have been developed both in EU and globally. These include the Europe's Beating Cancer Plan[17], the EU Strategy to support Member States in reducing alcohol related harm[18], the WHO Global Action Plan for the prevention and control of non-communicable diseases 2013-2020[19] and it updated annex 3[20], as well as the Global Alcohol Action Plan 2022-2030[21] put forward policy options to reduce the harmful use of alcohol (European Commission, 2024). For these reasons, the inclusion of **Alcohol consumption** as an indicator, also recognized under the Sustainable Development Goals (SDGs), is relevant. The corresponding indicator measures the amount of alcohol consumed by adults (15+ years) over a calendar year, expressed (World Health Organization, 2023).

Diets low in wholegrains are a major risk factor for disease burden in the EU. Recommendations for achieving sustainable, healthy diets emphasize the importance of including wholegrains in the daily diet. **Wholegrains consumption** is estimated by the Global Dietary Database[22]. However, due to the different data source, this indicator cannot currently be shown together with other food consumption estimates[23]. This may change in the future if an appropriate technical solution is developed.

3.4.2 Future development

As shown in **Table 11**, we plan to increase the number of indicators in the Social dimensions across all thematic areas. Additionally, some domains within the Fair, inclusive and ethical food system and the Food environment thematic areas will be populated with indicators, making these domains visible in the dashboard. Specifically, these are the **Social protection and poverty** domain, within the Fair, inclusive and ethical food system, thematic area, and the Food messaging and Properties of food domains within the Food environment thematic area.

Table 11. Overview of expected placeholder indicator development in the Social dimension

Thematic area	Domain	Placeholder indicator	Work planned
Fair, inclusive and ethical food system	Employment	Informal employment in agriculture	New area of research. Extract data from ILO, related to UN SDG indicator 8.3.1 and calculate the share of informal employment in agriculture.

17 https://health.ec.europa.eu/document/download/26fc415a-1f28-4f5b-9bfa-54ea8bc32a3a_en

18 https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=legissum:c11564b

19 https://www.who.int/publications/i/item/9789241506236

20 https://cdn.who.int/media/docs/default-source/ncds/mnd/2022-app3-technical-annex-v26jan2023.pdf

21 https://www.who.int/teams/mental-health-and-substance-use/alcohol-drugs-and-addictive-behaviours/alcohol/our-activities/towards-and-action-plan-on-alcohol

22 https://globaldietarydatabase.org/

23 The current specifications of our IT system allow only one value for data source metadata element.

Thematic area	Domain	Placeholder indicator	Work planned
		EU seasonal workers in agriculture	New area of research. Exploratory work ongoing, lack of data and data gaps in the available sources to evaluate.
		Non-EU seasonal workers in agriculture	New area of research. Exploratory work ongoing, data gaps to evaluate.
		Women employed in fisheries	Assessment of data collected within the forthcoming Annual Economic Reports.
	Social protection and poverty	Inability to afford a meal every second day	Data from private households and individuals living in the national territories.
		In work at-risk-of-poverty rate, in agri-food sector	New area of research. Exploratory work ongoing, the information on the prevalent sector of activity is not available yet.
	Animal welfare		Exploratory work ongoing.
Food environment	Food messaging	Food labelling	Indicators on FOP and sustainability related labelling to be explore and developed using MINTEL data.
	Properties of food	Nutritional quality of food offer	Ongoing work using food composition data on food and beverages from FABLE[24]
Nutrition and health	Health impact from diets	Prevalence of overweight and obesity among children	Update of childhood obesity indicators, with WHO data

Source: Own elaboration

3.5 Indicators of the horizontal thematic areas

Work on the conceptualisation of assessing good system governance and its resilience is ongoing. The presentation of the assessment framework, the methods for selecting and calculating the indicators will be the subject of the next update reports.

[24] https://food-labels-explorer.jrc.ec.europa.eu/en

4 Health check of the system

To enhance users' experience and the usability of the dashboard, parsimony ([25]) of the monitoring framework should be achieved, which can be approached in two ways:

— Discarding highly correlated indicators and

— Using composite indicator to summarise existing indicators.

These steps can be implemented after a rigorous statistical analysis of the available data, which prevents losing information while suggesting possible interventions. For this purpose, a periodic review of the indicators selected for the dashboard is planned, in accordance with the guide of the JRC's Competence Centre on Composite Indicators and Scoreboards (COIN).

The following subsections cover the COIN's methodology applied to the EU FSMF. The analysis was performed using the values of headline indicators as they appeared in the system on 18th of March 2025. The evolution of the dashboard (i.e. with the additions of new indicators and new data points) requires periodic repetitions of this analysis – also known as health checks of the system.

The COIN has developed a comprehensive guide for creating composite indicators, which consists of ten steps. So far, the creation of the FS dashboard covered steps 1 – 3, which include, defining the concept and objectives, selecting the component indicators, and collecting and pre-processing the data, and to some extent, assessment of data quality in terms of geographic and temporal coverage. In this section, the data and indicators are assessed for their suitability to create one or more composite indicators using the guidelines. It should however be noted that expert knowledge and substantive considerations may augment these general guidelines that are mainly based on statistical considerations. The primary method of assessment is Principal Component Analysis (PCA), which is presented at the end of the section. First, a brief review of the guidelines and the data preparation for the analysis is explained.

4.1.1 Data preparation

Any data analysis is only as good as the quality of the underlying data. The assessment of data quality includes timeliness, availability, and its distributional properties. The guidelines require the data to be:

— Recent – collected within the last five years,

— Available for at least 75% of the lowest units (indicator or sub-indicator),

— Available for 66% of indicators within each aggregation unit and

— Approximately normally distributed without outliers.

Concerning the timeliness for this preliminary analysis, any data that was older than 8 years was excluded. Additionally, only the last available year of data was used for the analysis, i.e. the analysis does not include time-trends. Using the first three criteria, a number of indicators were excluded

([25]) Representation or a model that uses relatively few independent variables to obtain a good fit to the data.

from the analysis as summarised in **Table 12**. There are few indicators dropped due to these restrictions. The exceptions are the "Trade" domain which included data on a wide range of products but only for the EU27 as a whole. The "Food Heritage" domain includes data on specific products under national protection schemes, which are not present in all EU27 MS.

Table 12. Number of sub-indicators excluded from the analysis due to the timeliness of data

Dimension	Area	Domain	N
Environmental	Biodiversity	Biodiversity conservation and restoration of natural ecosystems	1
	Cross-cutting environmental	Circular economy	1
		Food loss and waste	3
	Pollution and antimicrobials	Pollution	3
	Sustainable use of resources	Aquatic living resources	2
Economic	Development and logistics	Transport, accessibility and infrastructure	2
	Fair economic viability in food value chain	Price	1
		Trade	39
Social	Fair, inclusive and ethical food system	Employment	1
	Food environment	Food heritage	31
	Nutrition and health	Health impact from diets	1

NB. Sub-indicators are the lowest units in the analysis

Source: Own elaboration

In course of the analysis, one strategy to deal with missing data that remains after excluding variables is through imputation. Appropriate methods, such as mean or median imputation, regression imputation, or k-nearest neighbours (k-NN) could be implemented to alleviate missing data issues. The choice of method depends on the nature of the data and the percentage of missing data, and expert considerations. Often implementing different methods for each indicator is the best strategy, as it can incorporate the most knowledge and best practices. For the Principal Component Analysis presented here, the k-NN method was used on the entire dataset using three nearest neighbours.

Outliers can unduly influence aggregation and prevent other indicators from contributing to aggregate scores. Thus, the data were scanned for outliers using statistical methods. The COIN guidelines are to treat any indicators, which have an absolute skew > 2 and a kurtosis > 3.5. To treat outliers,

developers can consider either removing them, transforming the data, or ignoring them. In preparation for the PCA presented here, the first outlier was winsorised[26]. If the distribution did not pass the outlier test after winsorisation, then the original distribution was logged. **Table 13** below shows the number of indicators that were winsorised (win.) or logged (log.).

Table 13. Number of units (indicators or sub-indicators) treated using winsorisation or logarithm due to outliers

Dimension	Area	Domain	Win.	Log.
Environmental	Cross-cutting environmental	Circular economy	4	6
		Food loss and waste	3	1
	Sustainable use of resources	Land and soil	1	-
		Water	1	-
	Biodiversity	Genetic biodiversity of food production systems	-	2
	Climate change	GHG emissions	-	1
	Pollution and antimicrobials	Pollution	-	1
	Sustainable use of resources	Energy	-	3
Economic	Development and logistics	Transport, accessibility and infrastructure	2	-
	Fair economic viability in food value chain	Sectoral growth	1	-
		Trade	2	9
		Price	-	7
Social	Food environment	Food heritage	1	-
	Fair, inclusive and ethical food system	Animal welfare	-	1
	Food environment	Food affordability	-	1
		Food availability	-	1
	Nutrition and health	Nutrition and healthy, sustainable diets	-	1

Source: Own elaboration

When aggregating data, each indicator should be able to contribute to the final score i.e. have a similar range and variance. To ensure this, the data is normalised. For the preliminary analysis all indicators were mapped to the range of 1-100. Additionally, data must reflect intensive properties of a system and thus be denominated. Any variables not denominated are excluded from the analysis.

4.1.2 Principal component analysis

The PCA takes as input several indicators and assesses to what extent these can be combined into a few principal components without losing too much information. The resulting principal components

[26] Replacing an outliner with a value that is closer to the centre of the distribution.

should capture at least 75% of the information. The amount of information is represented by variance. A second condition is that any one principal component should capture at least one unit of variance, which is equivalent to the unique information one indicator provides to the pool of information that the PCA attempts to capture. The quantity of information that a principal component captures is represented by its eigenvalue, which should be greater than one. Thus, a collection of indicators is summarised successfully by a PCA if there are one or more principal components where each of them has an eigenvalue of at least one and cumulatively capture at least 75% of the variance.

In the analysis below, each indicator within a dimension of the Food System dashboard was involved in a PCA, and its components were extracted up to the number of domains within the dimension. The assumption was that domains will be the first level of aggregation for the indicators. In some cases, however, the dashboard contains more levels of aggregation, which means that an indicator may include sub-indicators that may constitute additional aggregation levels. Such levels should be scrutinised in future research. Nevertheless, the current analysis will provide a high-level overview of possibilities using the current set of indicators.

The environmental dimension includes eight domains, the economic dimension seven domains, and the social dimension 11 domains. Note that the number of domains may differ from those in the dashboard, as these are the remaining domains after excluding indicators due to data quality criteria mentioned above.

Table 14 shows the results of the PCA on the **Environmental dimension**. The PCA suggests that the environmental dimension could be successfully aggregated using six components, where each have an eigenvalue of at least one and together capture at least 75% of the variance. The number of components needed to reach 75% of the cumulative variance is two less than the number of domains (8).

The PCA also shows that increasing the number of domains even further is also possible. The PCA extracts up to 11 components from the Environmental dimension, which would capture 90% of the variance of the indicators. Additionally, the PCA shows that the first and second components capture 46% of the variance together. This indicates that a large number of items can be summarised by only two components. Examining PCA of indicators within each domain shows that the Consumption Footprint domain includes highly correlated indicators – some of which could be redundant. The GHG Emissions domain includes three negatively correlated indicators which should not be aggregated. Other domains contain one or two components, which could capture most of the variance. In conclusion, there are several ways aggregates can be created using Environmental indicators.

Table 14. Principal component analysis of indicators in the Environmental dimension

Principal component	Eigenvalue	Percentage of variance	Cumulative % of variance
1	23.45	34.49	34.49
2	8.41	12.37	46.86
3	7.04	10.35	57.21
4	5.96	8.77	65.97
5	4.49	6.60	72.58
6	3.26	4.80	77.37
7	2.20	3.23	80.61
8	2.01	2.95	83.56

Principal component	Eigenvalue	Percentage of variance	Cumulative % of variance
9	1.75	2.57	86.13
10	1.52	2.23	88.37
11	1.39	2.05	90.42

Source: Own elaboration

Table 15 shows the results of the PCA on the **Economic** dimension. It suggests that the indicators in this dimension could be successfully aggregated using 10 components, where each have an eigenvalue of at least one and together capture at least 75% of the variance. The number of components needed to reach 75% of the cumulative variance is three more than the number of domains (7) in the economic dimension. Thus, splitting domains is statistically advisable.

Principal component	Eigenvalue	Percentage of variance	Cumulative % of variance
1	16.93	18.21	18.21
2	11.36	12.22	30.42
3	8.18	8.80	39.22
4	7.21	7.75	46.97
5	6.37	6.85	53.82
6	4.87	5.24	59.06
7	4.64	4.99	64.05
8	4.14	4.46	68.50
9	3.60	3.87	72.37
10	3.17	3.40	75.78
11	2.61	2.81	78.59
12	2.31	2.48	81.07
13	2.14	2.30	83.37
14	2.01	2.16	85.53
15	1.91	2.05	87.59
16	1.62	1.74	89.33
17	1.60	1.72	91.04
18	1.43	1.54	92.58
19	1.31	1.41	93.99
20	1.23	1.33	95.31

Table 15. Principal component analysis of indicators in the Economic dimension

Source: Own elaboration

Moreover, the PCA shows increasing the number of domains even further is also possible as it extracts up to 20 components from the Economic indicators, which would capture 95% of the variance of the indicators. Additionally, the first and second components capture 30% of the variance together, which indicates that a large number of items could be summarised by only two components. Examining PCA of indicators within each domain we can conclude that all domains have two

or more components, except for the Price domain, which has only one. Therefore, if composite indicators were created it would be advisable to calculate sub-indexes within domains to maximise the variation captured.

Table 16 summarises the results of the PCA on the **Social** dimension. The PCA suggests the social dimension can be successfully aggregated using nine components, where each have an eigenvalue of at least one and together capture at least 75% of the variance. The number of components needed to reach 75% of the cumulative variance is one more than the number of domains (8). Thus, splitting domains is statistically possible and advisable.

Moreover, the PCA indicates that increasing the number of domains even further is also possible. The PCA extracts up to 13 components from the social dimension, which would capture 88% of the variance of the indicators. Therefore, the PCA shows the social dimension indicators could be summarised in different ways, depending on expert considerations and user needs.

Table 16. Principal component analysis of indicators in the Social dimension

Principal component	eigenvalue	Percentage of variance	Cumulative % of variance
1	9.08	20.19	20.19
2	7.04	15.63	35.82
3	4.07	9.04	44.86
4	3.54	7.87	52.73
5	2.89	6.42	59.15
6	2.16	4.79	63.94
7	2.04	4.53	68.47
8	1.86	4.14	72.61
9	1.84	4.09	76.70
10	1.71	3.81	80.51
11	1.38	3.06	83.57
12	1.12	2.48	86.05
13	1.05	2.34	88.39

Source: Own elaboration

4.1.3 Current health and future directions

The PCA has shown that each dimension of the dashboard is statistically coherent, with most of the indicator collections giving a nuanced yet coherent overview of the described phenomena. Additionally, the PCA indicates that the dashboard indicators could be aggregated. However, the data limitations described in Section 2.1 must be addressed before composite indicators can be created by aggregating the indicators. Expert inputs must provide the basis for the most appropriate way to handle missing data, select indicators which meet the quality standards, and treatment of outliers.

5 Conclusion

To address the identified knowledge gaps in the EU Food System Monitoring Framework (FSMF), a systematic effort was launched to develop the placeholders. This work included, on one hand, the maintenance of the placeholder list to incorporate the results of new research and to respond to the requirements of the evolving policy landscape. On the other hand, the identified placeholders were implemented by carrying out the required methodological developments and/or data collection. The development of placeholders is an ongoing task, planned until 2026."

The 15 new indicators represent a positive step towards achieving even indicator coverage across sustainability dimensions and food supply chain components, particularly improving the Economic and Social dimensions. However, our analysis shows that subject-specific indicators related to the middle of the supply chain, especially for Food distribution, are still lacking, highlighting the direction for further efforts. This work is closely aligned with current EU policies, such as the European Green Deal and the Vision for the future of agriculture and food, which emphasize the need for a sustainable and resilient food system and will contribute to the monitoring and evaluation of progress towards these policy goals.

The next phase of work will prioritize implementing the remaining placeholders and addressing areas for improvement, as identified through statistical analysis of the totality of data, to enhance the overall equity of the dashboard, with the ultimate goal of achieving a comprehensive and balanced monitoring framework.

References

Acs, S., Costa Leite, J., Sanyé-Mengual, E., Caivano, A., Catarino, R., et al., 'Towards sustainable food systems: developing a monitoring framework for the EU', *Frontiers in Sustainable Food Systems*, Vol. 8, 2024, https://doi.org/10.3389/fsufs.2024.1502081

Chen, X., Du, J., Wu, X., Cao, W., & Sun, S., 'Global burden attributable to high sodium intake from 1990 to 2019', *Nutrition, Metabolism and Cardiovascular Diseases*, Vol. 31(12), 2021, pp. 3314–3321, https://doi.org/https://doi.org/10.1016/j.numecd.2021.08.033

De Laurentiis, V., Biganzoli, F., Valenzano, A., & Sala, S., *Estimating food waste generated and packaging placed on the market at national level*. Publications Office of the European Union, Luxembourg, 2025, https://doi.org/10.2760/21595

De Laurentiis, V., Patinha Caldeira, C., Biganzoli, F., & Sala, S., *Building a balancing system for food waste accounting at national level*. Publications Office of the European Union, Luxembourg, 2021, https://doi.org/10.2760/316306

EFSA, 'Opinion of the Scientific Panel on Dietetic products, nutrition and allergies [NDA] related to the Tolerable Upper Intake Level of Sodium', *EFSA Journal*, 2005, https://doi.org/10.2903/j.efsa.2005.209

European Commission, Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. A Farm to Fork Strategy for a fair, healthy and environmentally-friendly food system. COM/2020/38. *Official Journal of the European Union,* 2020, https://eur-lex.europa.eu/legal-content/EN/LSU/?uri=celex:52020DC0381

European Commission, Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. Report on the state of the Digital Decade 2023. *Official Journal of the European Union,* 2023, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52023DC0570

European Commission, 'Risk factors for non-communicable diseases', 2024, European Commission website, 01 October 2024, accessed on 5 November 2025, https://knowledge4policy.ec.europa.eu/health-promotion-knowledge-gateway/topic/risk-factors-non-communicable-diseases_en

European Commission, Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. A Vision for Agriculture and Food Shaping together an attractive farming and agri-food sector for future generations, *Official Journal of the European Union,* 2025, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025DC0075.

Garcia Herrero, L., Casonato, C., Perez Cornago, A., Sanye Mengual, E., Sarasa Renedo, A., et al. *Overview and analysis of sustainable product procurement criteria in the EU food sector*. Publications Office of the European Union. Luxembourg, 2024 https://doi.org/doi/10.2760/1286793

GBD, 'GBD Results', Institute of Health Metrics and Evaluation (IHME) website, 2023, accessed 5 November 2025. https://vizhub.healthdata.org/gbd-results

Lasarte Lopez, J., De Jong, B., Gurria, P. and M`Barek, R, *Indicators for monitoring income distribution in the bioeconomy and the food system*, Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/9545814.

Schebesta, H., & Candel, J. J. L., 'Game-changing potential of the EU's Farm to Fork Strategy', *Nature Food*, Vol. 1(10), 2020, pp. 586–588. https://doi.org/10.1038/s43016-020-00166-9

Springmann, M., Spajic, L., Clark, M. A., Poore, J., Herforth, A., et al., 'The healthiness and sustainability of national and global food based dietary guidelines: Modelling study', *The BMJ*, Vol. 370, 2020, https://doi.org/10.1136/bmj.m2322

Tóth, K., Ács, S., Aschberger, K., Barbero Vignola, G., Bopp, S., et al. (2024). *EU food system monitoring framework. From concepts to indicators*. Publication Office of the European Union, 2024, https://doi.org/10.2760/94456

UN SDSN, *Indicators and a Monitoring Framework for the Sustainable Development Goals Launching a data revolution for the SDGs. Report to the Secretary-General of the United Nations by the Leadership Council of the Sustainable Development Solutions Network,* 2025,. https://sdgs.un.org/sites/default/files/publications/2013150612-FINAL-SDSN-Indicator-Report1.pdf

World Health Organization. 'Alcohol, total per capita (15+) consumption (in litres of pure alcohol)', The Global Health Observatory website, 2025, accessed on 5 November 2025, https://www.who.int/data/gho/data/indicators/indicator-details/GHO/total-(recorded-unrecorded)-alcohol-per-capita-(15-)-consumption, (2023).

List of abbreviations and definitions

Abbreviations	Definitions
CEAP	Circular economy action plan
COIN	Competence Centre on Composite Indicators and Score-boards
DG	Directorate-General
DALY	Disability-adjusted life years
EC	European Commission
EEA	European Economic Area
EGD	European Green Deal
ESFA	European Food Safety Authority
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FMSY	Fishing mortality at maximum sustainable yield
FS	Food system
FSMF	Food system monitoring framework
FS-MFA	Food System Material Flow Analysis
GHG	Greenhouse gas
IoT	Internet of things
JRC	Joint Research Centre
k-NN	k-nearest neighbours
LCA	Life cycle assessment
NCDs	Non-communicable diseases
PCA	Principal component analysis
SDG	Sustainable development goals
WHO	WHO World Health Organization

List of figures

Figure 1. Food system sustainability model of the EU food system monitoring framework..... 6

Figure 2. Notation of changes in the dashboard..... 8

Figure 3. Proportions of the sustainability dimensions in the dashboard..... 10

Figure 4. Proportion of food supply chain components in the dashboard..... 11

Figure 5. Steps of placeholder processing..... 12

List of tables

Table 1. Evolution of indicator numbers within the sustainability dimensions of the EU food system monitoring dashboard 9

Table 2. Changes of indicator numbers within the supply chain components of the dashboard 11

Table 3. New and updated indicators elaborated during the first upgrade release of the EU Food System Monitoring Framework 13

Table 4. New and updated indicators for Cross-cutting environmental thematic area 15

Table 5. Overview of expected placeholder indicator development in the Environmental dimension 16

Table 6. New and updated indicators for Fair economic viability in food value chain 19

Table 7. New and updated indicators for Development and logistics 20

Table 8. Overview of expected placeholder indicator development in the Economic dimension 21

Table 9. New and updated indicators for Fair, inclusive and ethical food system 22

Table 10. New and updated indicators for Nutrition and health 24

Table 11. Overview of expected placeholder indicator development in the Social dimension 26

Table 12. Number of sub-indicators excluded from the analysis due to the timeliness of data 29

Table 13. Number of units (indicators or sub-indicators) treated using winsorisation or logarithm due to outliers 30

Table 14. Principal component analysis of indicators in the Environmental dimension 31

Table 15. Principal component analysis of indicators in the Economic dimension 32

Table 16. Principal component analysis of indicators in the Social dimension 33

Table 17. Comprehensive list of indicators in the EU food system monitoring dashboard 40

Annexes

Annex 1. Recapitulative list of all indicators in the EU food system monitoring dashboard

Table 17. Comprehensive list of indicators in the EU food system monitoring dashboard

Thematic areas	Domains	Indicator	Category
Environmental			
Climate change	GHG emissions	GHG food system emissions	H
		Fishing-related CO2 emissions related to fuel used per kg of landings (EU)	S
		GHG emissions from agriculture	S
		Net GHGs Emissions from LULUCF sector	S
Pollution and anti-microbials	Pollution	Use and risk of chemical pesticides (F2F pesticide reduction target 1)	H
		Use of the more hazardous pesticides (F2F pesticide reduction target 2)	H
		Water quality - Nitrates in groundwater	H
		Ammonia emissions from agriculture	S
	Antimicrobials	Sales of antimicrobials for food-producing animals	H
Sustainable use of resources	Land and soil	Gross nutrient balance - nitrogen	H
		Gross nutrient balance - phosphorus	H
		Share of agricultural area under organic farming	H
		Soil organic carbon in agricultural land	H
		Consumption of inorganic fertilizers - Nitrogen	S
		Consumption of inorganic fertilizers - Phosphorus	S
		Share of the top three crops in the total agricultural area	S
	Water	Water exploitation index, plus (WEI+)	H
	Aquatic living resources	Fishing pressure relative to maximum sustainable yield	H
		Fish stock biomass relative to biomass in 2003	S
	Energy	Final energy consumption in agriculture, forestry and the food industry	H
		Fuel use of fisheries per Kg of fish landed in ports	S

		Production of renewable energy from agriculture and forestry	S
Biodiversity	Biodiversity conservation and restoration of natural ecosystems	Common farmland birds indicator	H
		Consumption Footprint - Food (biodiversity loss)	H
		Grassland butterfly Index	S
	Genetic biodiversity of food production systems	Number of genetic resources for food production secured in either medium-or long-term conservation facilities	S
Cross-cutting environmental	Food loss and waste	Food losses	H
		Food waste	H
		Direct agricultural loss attributed to disasters	S
	Circular economy	Generation and use of by-products	H
	Consumption Footprint	Consumption - Footprint - Food	H
		Consumption Footprint - Food (acidification)	S
		Consumption Footprint - Food (climate change)	S
		Consumption Footprint - Food (eutrophication, freshwater)	S
		Consumption Footprint - Food (eutrophication, marine)	S
		Consumption Footprint - Food (eutrophication, terrestrial)	S
		Consumption Footprint - Food (freshwater ecotoxicity)	S
		Consumption Footprint - Food (human toxicity, cancer)	S
		Consumption Footprint - Food (Human toxicity, non-cancer)	S
		Consumption Footprint - Food (ionising radiation)	S
		Consumption Footprint - Food (land use)	S
		Consumption Footprint - Food (ozone depletion)	S
		Consumption Footprint – Food (particulate matter)	S
		Consumption Footprint - Food (Photochemical ozone formation)	S
		Consumption Footprint - Food (resource use, fossil)	S
		Consumption Footprint - Food (resource use, minerals and metals)	S

		Consumption Footprint - Food (water use)	S
Economic			
Fair economic viability in food value chain	Sectoral growth	Labour productivity of the different sectors of the food chain	H
		Value Added along the food chain	H
	Market power and business structure	Gross fixed capital formation in agriculture	H
		Fish landings of the EU small-scale fisheries	S
	Income distribution	Average salary by sector	H
		Farmers' income in agriculture compared to wages in the rest of the economy	H
		Employee earnings ratio	S
		Labour income share	S
	Price	Consumer food inflation	H
		Price indices of agricultural inputs	H
		Share of household spending on food	H
	Trade	Self-sufficiency rates - commodities	H
		Trade balance	H
Development and logistics	Technology and digitalisation	Digital intensity in food manufacturing enterprises	H
		Digital skills of the labour force in the agriculture, forestry or fishing sectors	H
		Agricultural training of farm managers	S
		Rural Next Generation Access broadband coverage	S
		Use of Artificial Intelligence in the food manufacturing sector	S
		Use of Internet of Things (IoT) in the food manufacturing sector	S
	Transport, accessibility and infrastructure	Annual road freight transport by distance class	H
Social			
Fair, inclusive and ethical food system	Employment	Employment by economic activity	H
		Number of fishers in the EU small-scale fisheries (passive gears)	H
		Young farm managers in agriculture	H
		Young fishers	H
		Accidents at work	S
		Gender employment gap in the food sector	S
		Job vacancies in the agrifood sector	S
	Animal welfare	Organic production of aquaculture products	H

		Share of laying hens by farming method	H
Food environment	Food heritage	Number of products under a quality scheme per category and per scheme	H
	Food affordability	Affordability of a healthy diet: ratio of cost to food expenditures	H
		Share of the population unable to afford a healthy diet	H
	Food availability	Ratio plant to total protein supply	H
		Supply by food group	H
Nutrition and health	Nutrition and healthy, sustainable diets	Adult consumption by food groups	H
		Alcohol consumption	H
		Salt intake	H
		Wholegrains consumption	H
	Health impact from diets	Burden of disease attributable to dietary risk factors	H
		Prevalence of overweight and obesity among adults	H
		Prevalence of overweight and obesity among children (aged 6 to 9 years)	H
		Prevalence of overweight and obesity among children (<5 years)	S
	Food security	Prevalence of moderate or severe food insecurity in the population	H

H– headline indicator, S– secondary indicator

Source: Own elaboration

Getting in touch with the EU

In person

All over the European Union there are hundreds of Europe Direct centres. You can find the address of the centre nearest you online (european-union.europa.eu/contact-eu/meet-us_en).

On the phone or in writing

Europe Direct is a service that answers your questions about the European Union. You can contact this service:

— by freephone: 00 800 6 7 8 9 10 11 (certain operators may charge for these calls),
— at the following standard number: +32 22999696,
— via the following form: european-union.europa.eu/contact-eu/write-us_en.

Finding information about the EU

Online

Information about the European Union in all the official languages of the EU is available on the Europa website (european-union.europa.eu).

EU publications

You can view or order EU publications at op.europa.eu/en/publications. Multiple copies of free publications can be obtained by contacting Europe Direct or your local documentation centre (european-union.europa.eu/contact-eu/meet-us_en).

EU law and related documents

For access to legal information from the EU, including all EU law since 1951 in all the official language versions, go to EUR-Lex (eur-lex.europa.eu).

EU open data

The portal data.europa.eu provides access to open datasets from the EU institutions, bodies and agencies. These can be downloaded and reused for free, for both commercial and non-commercial purposes. The portal also provides access to a wealth of datasets from European countries.

Science for policy

The Joint Research Centre (JRC) provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society

Scan the QR code to visit:

The Joint Research Centre: EU Science Hub
https://joint-research-centre.ec.europa.eu

Publications Office
of the European Union